Table 1: Detailed per-seed results for the state consistency stress test (underdamped pendulum). Ground truth: $(\gamma_0, \gamma_1) = (4.0016, 0.08)$. The baseline (no nuisance) estimate from clean videos is $\hat{\gamma}_0 = 4.0081 \pm 0.0003$ and $\hat{\gamma}_1 = 0.0788 \pm 0.0005$. Entries in red indicate individual runs that deviated noticeably from the ground truth. The rightmost columns report the mean ± std across the 5 seeds. For Brightness Level 2, one of the five seeds converged to a poor local optimum; excluding it, the remaining four seeds yield $\hat{\gamma}_0 = 4.0070 \pm 0.0114$ and $\hat{\gamma}_1 = 0.0680 \pm 0.0111$. See Figure 1 for sample frames (seed 1 shown).

		Seed 1		Seed 2		Seed 3		Seed 4		Seed 5		Mean ± Std	
Category	Level	$\hat{\gamma}_0$	$\hat{\gamma}_1$	$\hat{\gamma}_0$	$\hat{\gamma}_1$	$\hat{\gamma}_0$	$\hat{\gamma}_1$	$\hat{\gamma}_0$	$\hat{\gamma}_1$	$\hat{\gamma}_0$	$\hat{\gamma}_1$	$\hat{\gamma}_0$	$\hat{\gamma}_1$
Bg-Noise	L1	4.0090	0.0785	4.0084	0.0815	4.0033	0.0796	4.0022	0.0801	4.0026	0.0804	4.0051 ± 0.0033	0.0800 ± 0.0011
	L2	4.0025	0.0807	4.0029	0.0811	4.0081	0.0811	4.0077	0.0792	4.0029	0.0805	4.0048 ± 0.0028	0.0805 ± 0.0008
	L3	4.0039	0.0813	4.0039	0.0806	4.0028	0.0813	4.0049	0.0810	4.0042	0.0817	4.0039 ± 0.0008	0.0812 ± 0.0004
Mv-Clutter	L1	4.0072	0.0798	4.0112	0.0769	4.0061	0.0793	4.0070	0.0796	4.0132	0.0790	4.0089 ± 0.0031	0.0789 ± 0.0012
	L2	4.0082	0.0812	3.9988	0.0801	3.9926	0.0871	4.0136	0.0768	4.0137	0.0696	4.0054 ± 0.0094	0.0790 ± 0.0064
	L3	4.0011	0.0559	2.4322	−2.6736	3.8276	−0.1967	3.9565	−0.2040	2.9499	0.1525	3.4335 ± 0.7049	−0.5732 ± 1.1845
Cam-Jitter	L1	4.0025	0.0801	4.0024	0.0807	4.0030	0.0796	4.0019	0.0794	4.0021	0.0796	4.0024 ± 0.0004	0.0799 ± 0.0005
	L2	4.0034	0.0803	4.0022	0.0799	4.0026	0.0794	4.0053	0.0798	4.0020	0.0798	4.0031 ± 0.0013	0.0798 ± 0.0003
	L3	4.0021	0.0805	4.0031	0.0799	4.0020	0.0795	4.0031	0.0804	4.0025	0.0783	4.0026 ± 0.0005	0.0797 ± 0.0009
Brightness	L1	4.0116	0.0776	4.0107	0.0788	4.0108	0.0779	4.0088	0.0782	4.0090	0.0773	4.0102 ± 0.0012	0.0780 ± 0.0006
	L2	3.4615	−3.0378	4.0093	0.0521	3.9902	0.0758	4.0140	0.0684	4.0145	0.0755	3.8979 ± 0.2442	−0.5532 ± 1.3890
	L3	4.0228	0.0632	3.9969	0.0404	4.0256	0.0441	4.0050	0.0444	4.0020	0.0011	4.0105 ± 0.0129	0.0386 ± 0.0228

Table 2: Real-video pendulum: string-length recovery. For each ground-truth length, we report the original string-length estimate (mean±std over 5 clips) and the corresponding RMSE (m).

	$L^\star = 0.45$ m		$L^\star = 0.90$ m		$L^\star = 1.50$ m	
Method	Estimate	RMSE	Estimate	RMSE	Estimate	RMSE
PAIG	1.01 ± 0.03	0.561	1.01 ± 0.04	0.116	1.01 ± 0.04	0.491
NIRPI	0.77 ± 0.33	0.435	0.84 ± 0.53	0.478	0.63 ± 0.38	0.934
LPFV	0.51 ± 0.01	0.061	1.07 ± 0.20	0.247	1.30 ± 0.02	0.201
Ours	0.50 ± 0.001	**0.050**	0.97 ± 0.003	**0.070**	1.60 ± 0.015	**0.101**

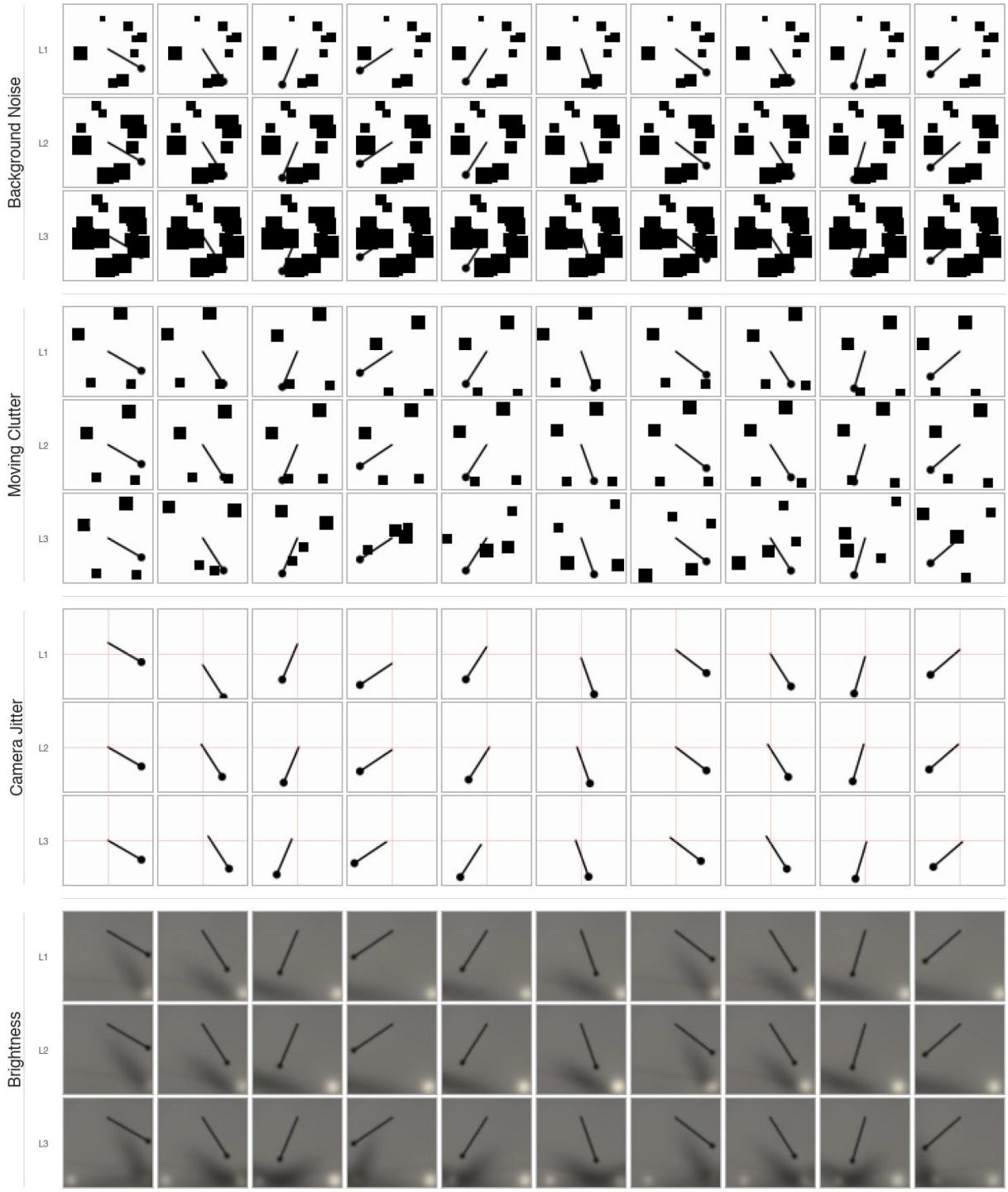

Figure 1: State consistency stress test: sample frames from the ablation videos (10 equally spaced frames from $t = 0$ to $t = 1.5\pi$; seed 1 shown). Each row corresponds to one severity level (L1–L3) of a visual nuisance category. The red crosshairs overlaid on the Camera Jitter rows are visualization-only reference lines (not present in the actual training videos) that mark the original frame center, making it easier to see that the pendulum is no longer centered after the simulated camera shake. **Background Noise:** static black squares of random sizes are overlaid on the background. L1: 8 squares (4–10 px); L2: 12 squares (6–14 px); L3: 16 squares (7–16 px). **Moving Clutter:** 4 black squares (4–10 px) move as distractors. L1: all squares follow a coordinated sinusoidal trajectory (± 10 px horizontal, ± 6 px vertical); L2: each square moves independently along its own smooth sinusoidal path (± 3 px horizontal, ± 1.6 px vertical); L3: each square bounces independently in random directions at 1.6 px/frame. **Camera Jitter:** the entire frame is shifted to simulate camera shake (exposed regions filled white). L1: smooth vertical oscillation (± 8 px, 7 full cycles); L2: 2D random walk (max $\pm 3/\pm 2$ px, 72% step probability, 45% directional momentum); L3: aggressive 2D random walk (max $\pm 5/\pm 4$ px, 88% step probability) with 18% chance of sudden 2–4 px burst jumps. **Brightness:** the scene is rendered with pseudo-3D lighting. L1: 1 light source with fixed position and fixed intensity; L2: 1 light source whose position and intensity both vary smoothly over time; L3: 2 independent light sources, each with smoothly varying position and intensity.

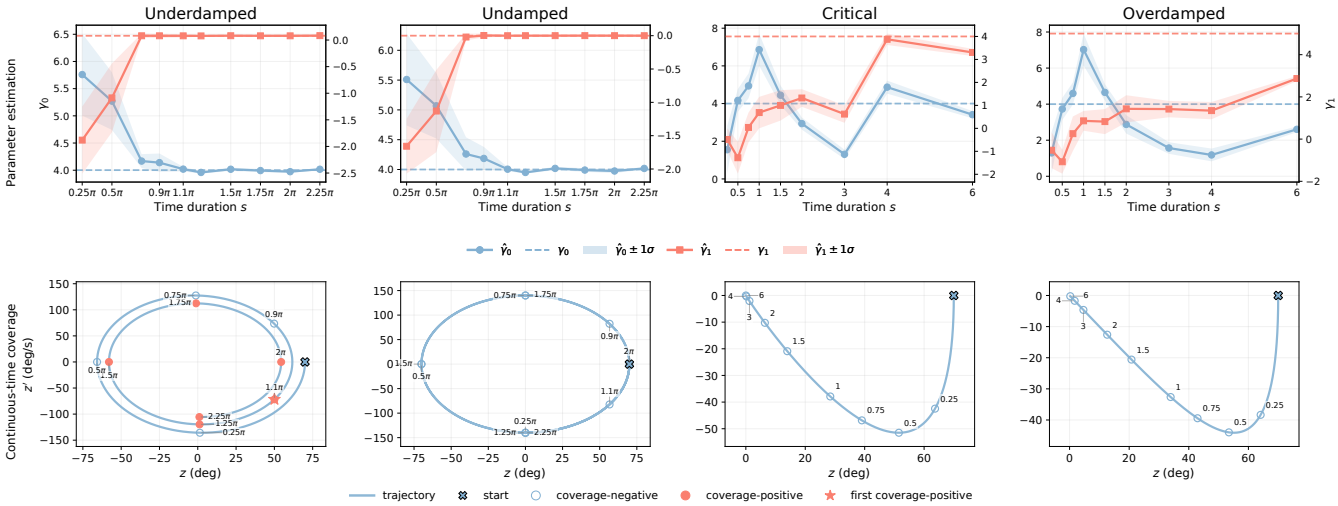

Figure 2: Single-trajectory pendulum results across four damping regimes. Top: recovered parameters versus tested video length (mean $\pm 1\sigma$ over 5 seeds; dashed lines show the ground-truth values). The left-side γ_0 and right-side γ_1 tick values are shown in all four panels for readability. Bottom: phase portraits of the true continuous-time state trajectory in the (z, z') plane for the same initial condition. The blue curve traces $(z(t), z'(t))$ from $t = 0$. The "x" marker denotes the start of the trajectory. Each labeled endpoint corresponds to one tested video length. Hollow blue circles denote coverage-negative endpoints, filled red circles denote coverage-positive endpoints, and the red star marks the first coverage-positive endpoint.

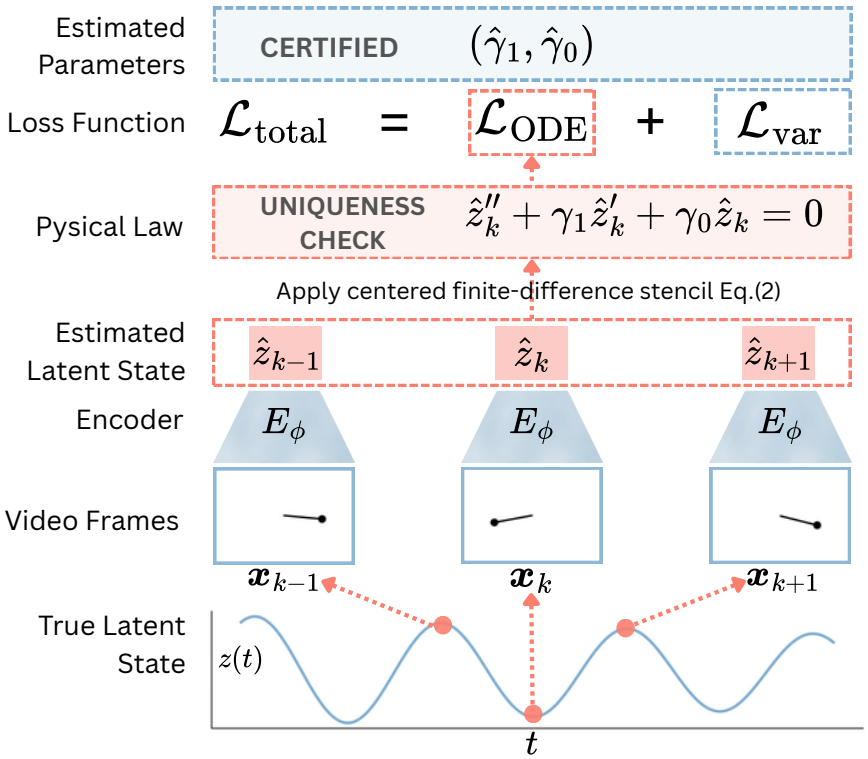

Figure 3: **Encoder-only pipeline with identifiability-aware parameter estimation.** A shared encoder E_ϕ maps video frames $\boldsymbol{x}_k$ to scalar latents $\hat{z}_k$. Using the centered finite-difference stencil, we compute $\hat{z}'_k$ and $\hat{z}''_k$ and enforce the second-order linear ODE with a residual loss $\mathcal{L}_{\text{ODE}}$. A variance-floor regularizer $\mathcal{L}_{\text{var}}$ prevents latent collapse and improves conditioning, yielding parameter estimates $(\hat{\gamma}_1, \hat{\gamma}_0)$. The *UNIQUENESS CHECK* box denotes the trajectory-level diagnostic: it tests whether the current clip/window satisfies the coverage condition required for uniqueness. The *CERTIFIED* label means that this diagnostic passes, so the recovered parameters are uniquely determined by the observed video, rather than merely achieving low residual loss.

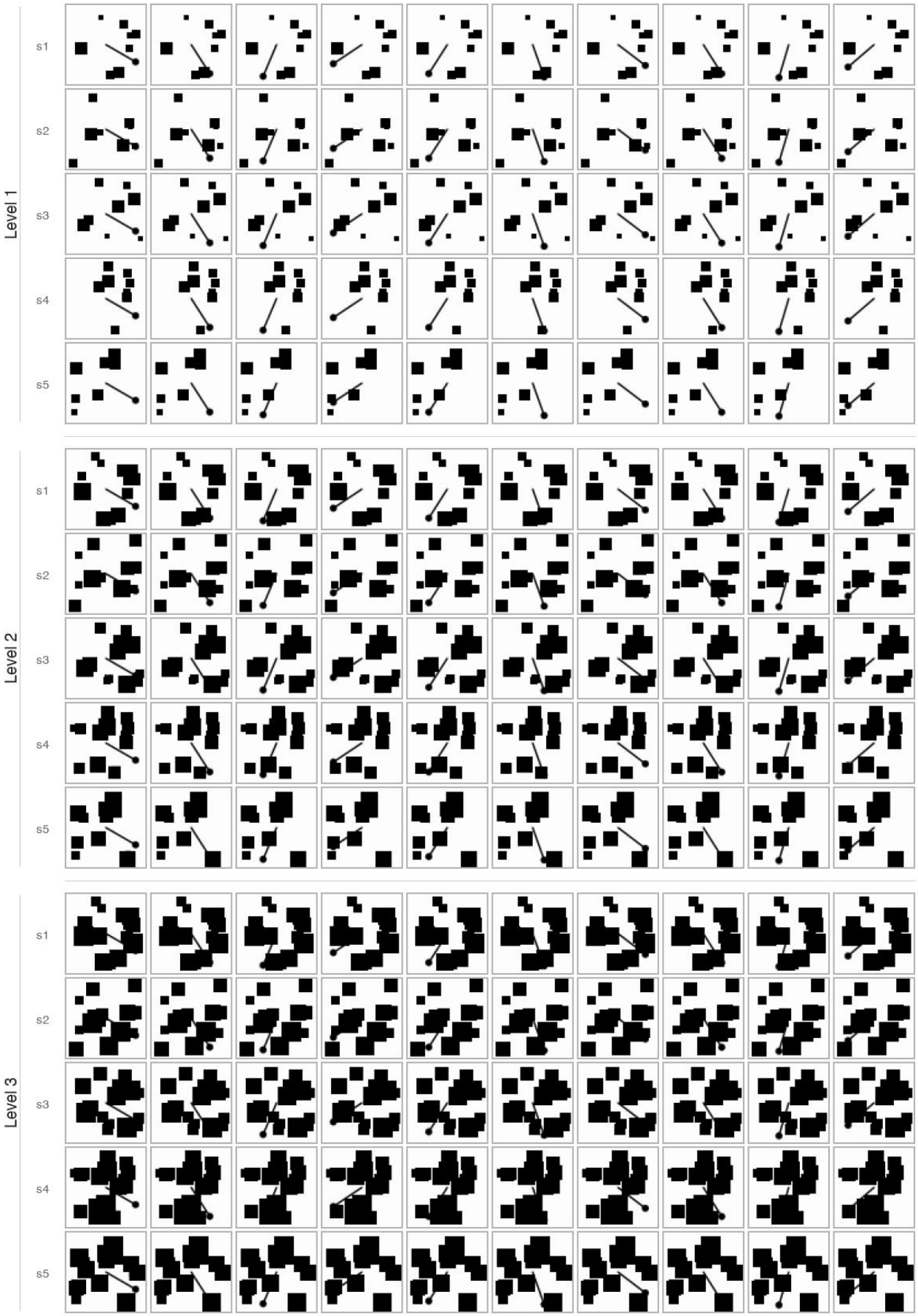

Figure 4: Background Noise ablation: all seeds (s1–s5) across Levels 1–3. See Figure 1 for level definitions.

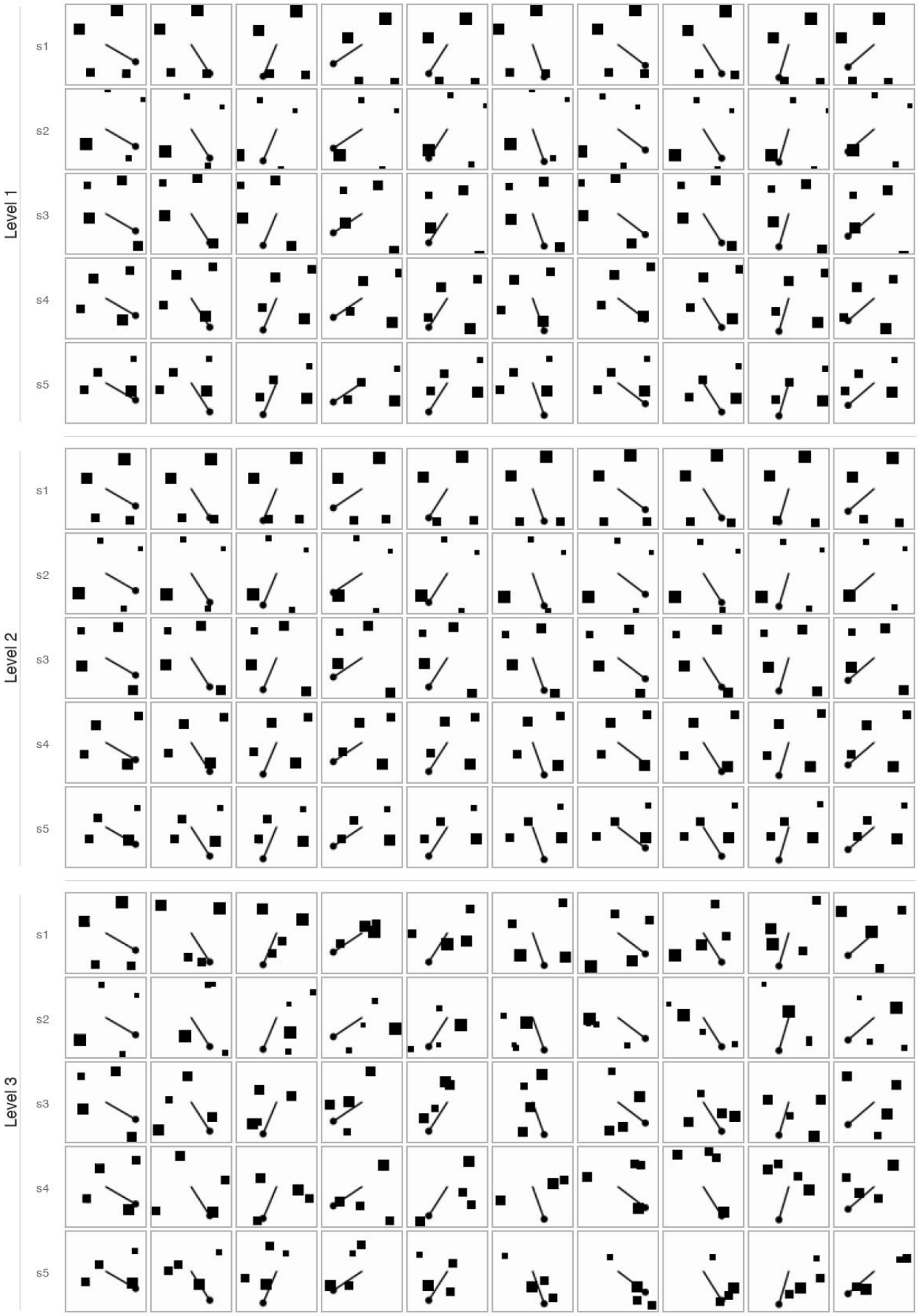

Figure 5: Moving Clutter ablation: all seeds (s1–s5) across Levels 1–3. See Figure 1 for level definitions.

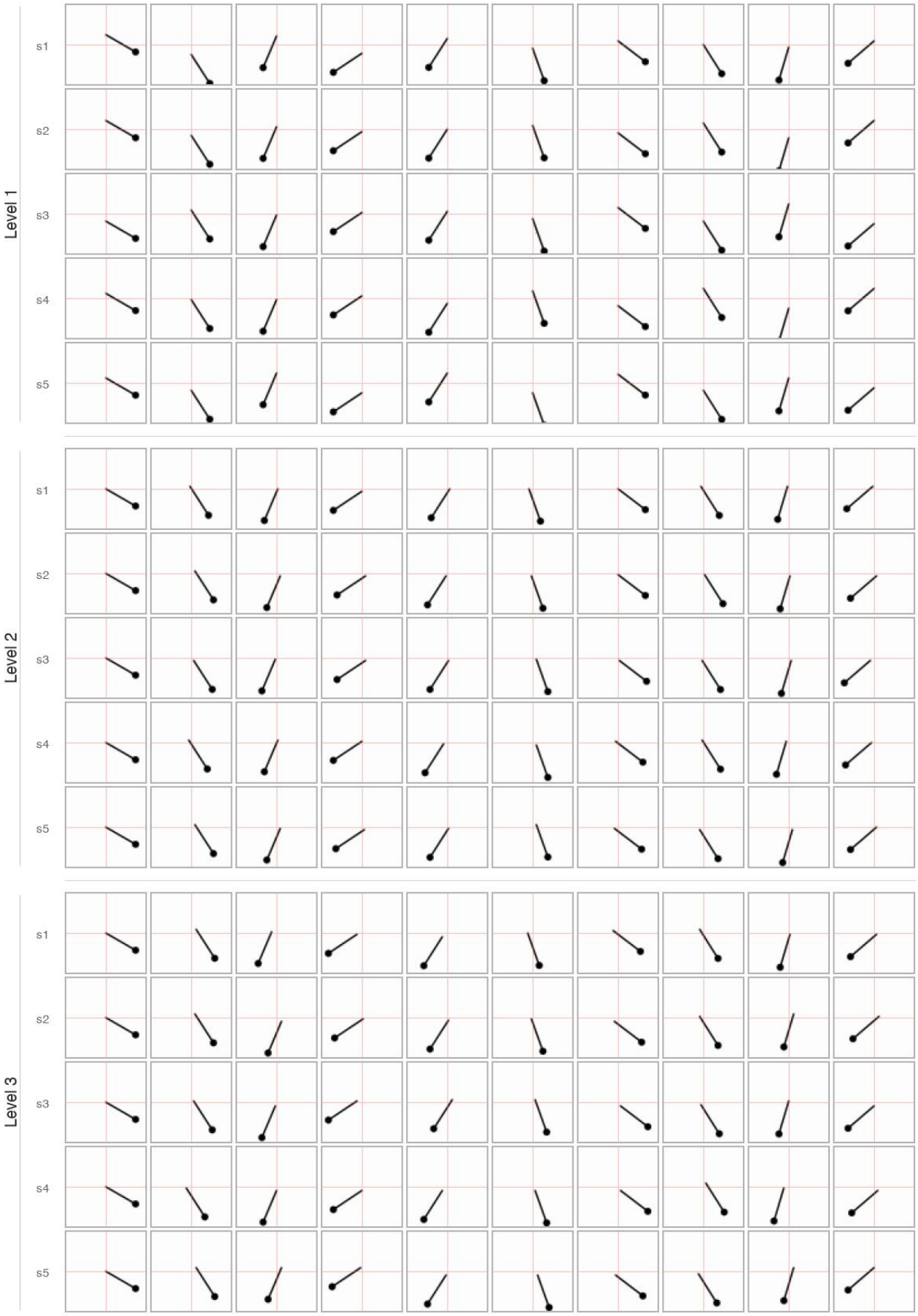

Figure 6: Camera Jitter ablation: all seeds (s1–s5) across Levels 1–3. See Figure 1 for level definitions.

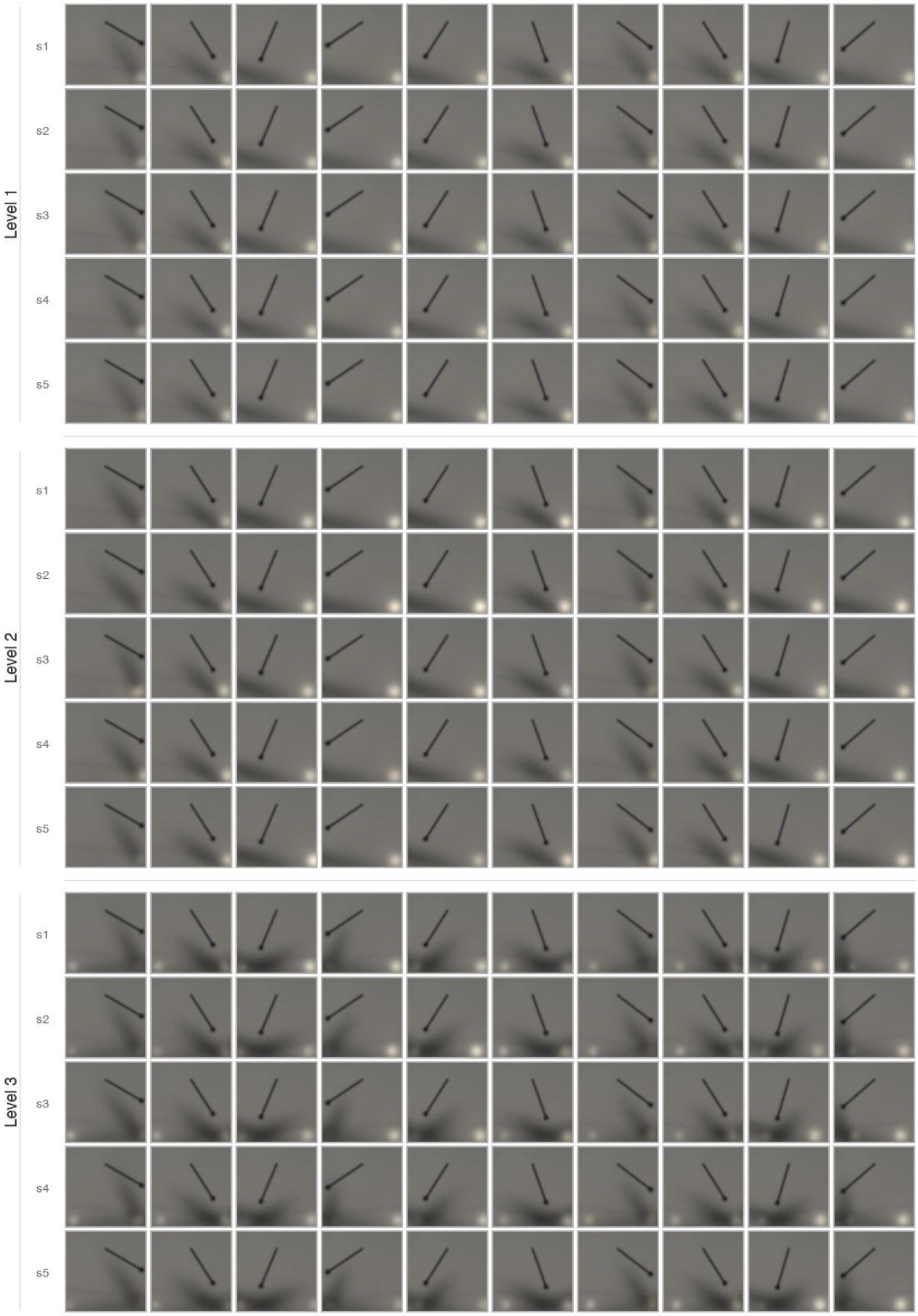

Figure 7: Brightness ablation: all seeds (s1–s5) across Levels 1–3. See Figure 1 for level definitions.